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Author

Topic: Falcon 9 R landing deceleration / Minimum Merlin th

☐ **speedevil**

Senior Member



Posts: 4406

Fife

Liked: 2762

Likes Given: 3368

[Falcon 9 R landing deceleration / Minimum Merlin throttle levels.](#)

« on: 01/06/2015 05:02 pm »

I've done some searching, and failed to find anything other than the general assumptions that the Falcon 9 R can throttle to 70% 'Stage weight about 25 tons'.

However.
The ORBCOMM



SpaceX



Watch on

video raises questions for me.

At 0:38 - 25:42 - the engine ignites.

At 26:06 - the engine is last seen.

At 70% of specified launch thrust, the thrust would be 455kN or so.
Or 18m/s^2 deceleration, with a 25000kg stage.
8m/s once you subtract gravity.

But - the 28 seconds of engine firing above implies 210m/s or so of delta-v, and the stage is under 100m/s as I understood the aerodynamics.

Dropping the specified launch thrust and going to the original engine design thrust and mu helps a little, but not much. (160m/s).

If Merlin can throttle to a little under 50% - the numbers work out.
Or have I dropped a zero somewhere?

☐ **cscott**

Senior Member



Posts: 3473

Liked: 2869

Likes Given: 726

Re: Falcon 9 R landing deceleration / Minimum Merlin throttle levels.

« **Reply #1 on:** 01/06/2015 05:08 pm »

"Throttle 70%" == total thrust 30% of maximum.

☐ **speedevil**

Senior Member



Posts: 4406

Fife

Liked: 2762

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Re: Falcon 9 R landing deceleration / Minimum Merlin throttle levels.

« **Reply #2 on:** 01/06/2015 05:10 pm »

That would, indeed do it.

But - <http://aviationweek.com/awin/spacex-unveils-plans-be-world-s-top-rocket-maker> -as i
"The 1D design incorporates many lessons learned from the earlier Merlins and is of a simp
an increased fatigue life. "We've added the ability to throttle between 70% and 100%. "

That seems to not leave the possibility that it means that.

« *Last Edit: 01/06/2015 05:12 pm by speedevil* »

☐ **cambrianera**

Full Member



Posts: 1438

Liked: 318

Likes Given: 261

Re: Falcon 9 R landing deceleration / Minimum Merlin throttle levels.

« **Reply #3 on:** 01/06/2015 05:55 pm »

FYI in the various Grasshopper thread minimum throttle was consistently calculated at 70%
last hop, when it was calculated at 60%.

Sometime after that Elon tweeted that the engine has been improved and can throttle 60%
This should be actual value, but some people believe 60% is the range and minimum value

 **cscott**, **hrissan** and **speedevil** like this

Oh to be young again. . .

☐ **speedevil**

Senior Member



Posts: 4406

Fife

Re: Falcon 9 R landing deceleration / Minimum Merlin throttle levels.

« **Reply #4 on:** 01/08/2015 08:36 pm »

As some context.

Liked: 2762
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This is the second stage oxygen tank.
Assuming that the base of the tank is hemispherical, and that it is full diameter - I make thi
so of the diameter of a hemisphere covered with fuel.
Or around a 30cm depth.
Of the order of 300l. (please note, this was a rough measurement in which I used a bowl I h
using for yogurt and a cracker. More accurate guesses using other foodstuffs welcome.)

Assuming it has the same burn duration - 28s as in the ORBCOMM, and the thrust level is 4
above) - and that the sea level impulse is 250 (10% down on max thrust), this is 2900kg of
3200L.

1250l of RP1, and 1900 of LOX.
Or in US gallons - 310 and 475.
To leave only 15 and 150 as discussed in the filings referred to in [http://forum.nasaspacefli
index.php?topic=36513.15](http://forum.nasaspacefli
index.php?topic=36513.15) is quite astounding.

It clearly can't be metered down from the ~400000l or so loaded into the tanks initially.

I guess in principle, you can measure to a few kilos - with differential pressure transducers
a lot of averaging on the 'smooth' portions of the descent into atmosphere.

 go4mars likes this

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